

Homework 4

Andrew Hwang

Math 38

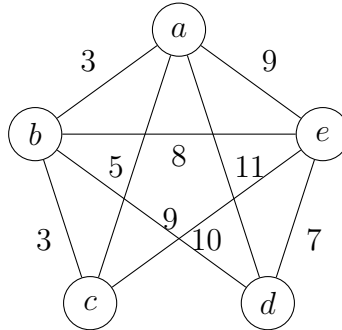
Gayee Park

April 29, 2026

Problem 1. There are five cities in a network. The cost of building a road directly between i and j is the entry $a_{i,j}$ in the matrix below. An infinite entry indicates that there is a mountain in the way and the road cannot be built. Determine the least cost of making all the cities reachable from each other.

$$\begin{pmatrix} 0 & 3 & 5 & 11 & 9 \\ 3 & 0 & 3 & 9 & 8 \\ 5 & 3 & 0 & \infty & 10 \\ 11 & 9 & \infty & 0 & 7 \\ 9 & 8 & 10 & 7 & 0 \end{pmatrix}$$

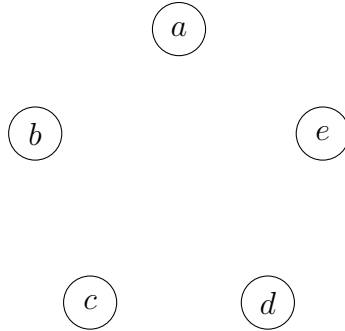
Solution. We first draw the graph as such based on the matrix given. Note that we denote the vertices a , b , c , d , and e . Because d and e have a weight of infinity, we simply do not draw an edge de :



Notice that in essence, this is a minimally spanning tree (MST) problem. Hence, let us perform Kruskal's algorithm. We are given the edges in non-decreasing order

$$[3, 3, 5, 7, 8, 9, 9, 10, 11].$$

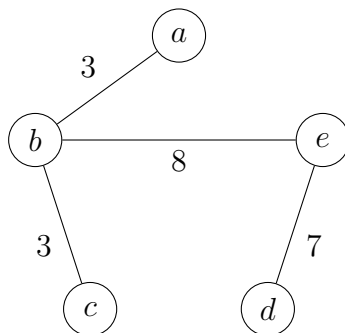
Consider a graph G' with 5 isolated vertices as shown below:



Let us start from the smallest weight. We are given the sequence as follows:

- Adding edge ab decreases the number of connected components, so we add ab to G' with $\text{wt}(ab) = 3$.
- Adding edge bc decreases the number of connected components, so we add bc to G' with $\text{wt}(bc) = 3$.
- Adding edge ac does not decrease the number of connected components, so we discard five from the list and move on.
- Adding edge de decreases the number of connected components, so we add de to G' with $\text{wt}(de) = 7$.
- Adding edge be decreases the number of connected components, so we add be to G' with $\text{wt}(be) = 8$.

The result is that the graph is connected, hence achieving the MST as shown with the total cost of $3 + 3 + 7 + 8 = \boxed{21}$.



□

Problem 2. Four people must cross a canyon at night on a fragile bridge. At most two people can be on the bridge at once. Crossing requires carrying a flashlight, and there is only one flashlight (which can cross only by being carried). Alone, the four people cross 10, 5, 2, 1 minutes, respectively. When two cross together, they move at the speed of the slower person. In 18 minutes, the bridge will collapse. Can the four people get across in time?

Solution. We will first denote the four people as A, B, C , and D , each with cross times of 10, 5, 2, and 1 minutes, respectively. First, let us start with all four people on the starting side. First, cross C and D , which costs two minutes. Then, bring back D , which costs one minute. Afterwards, cross A and B , which costs 10 minutes. Next, bring C back, which costs two minutes. Lastly, bring C and D back, which costs two minutes and completes the crossing. Adding the times up, we get

$$2 \text{ min} + 1 \text{ min} + 10 \text{ min} + 2 \text{ min} + 2 \text{ min} = 17 \text{ min} < 18 \text{ min}.$$

Hence, by following this specific procedure, all four people can cross the bridge in under 18 minutes. $\square$

Problem 3. Use the Matrix Tree Theorem to prove Cayley's Formula. Use the Matrix Tree Theorem to also compute the number of spanning trees for $K_{r,s}$.

Proof. Let us first find the Laplacian of a complete graph K_n . The degree of all vertices of a complete graph is $n - 1$, so B is a diagonal matrix all with $b_{ii} = n - 1$, and the adjacency matrix A is a $n \times n$ matrix with all non-diagonal entries 1 and all diagonal entries 0. Hence,

$$Q = B - A = L = \begin{pmatrix} n-1 & -1 & -1 & \cdots & -1 \\ -1 & n-1 & -1 & \cdots & -1 \\ -1 & -1 & n-1 & \cdots & -1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -1 & -1 & -1 & \cdots & n-1 \end{pmatrix}$$

Hence, by removing the first row and first column, the minor the matrix L^* is an $(n - 1) \times (n - 1)$ matrix

$$L^* = \begin{pmatrix} n-1 & -1 & -1 & \cdots & -1 \\ -1 & n-1 & -1 & \cdots & -1 \\ -1 & -1 & n-1 & \cdots & -1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -1 & -1 & -1 & \cdots & n-1 \end{pmatrix}$$

To find the determinant, let us use elementary row operations. First, let us operate on the first row by adding onto it all the other rows as such

$$L^* \xrightarrow{R_1 \rightarrow R_1 + R_2 + \cdots + R_{n-1}} \begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ -1 & n-1 & -1 & \cdots & -1 \\ -1 & -1 & n-1 & \cdots & -1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -1 & -1 & -1 & \cdots & n-1 \end{pmatrix} \xrightarrow{R_i \rightarrow R_i + R_1, i \geq 2} \begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ 0 & n & 0 & \cdots & 0 \\ 0 & 0 & n & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & n \end{pmatrix}$$

This is an upper triangular matrix, so the determinant is the products of the elements of the diagonals. Hence,

$$\tau(K_n) = \det(L^*) = 1 \cdot \underbrace{n \cdot \cdots \cdot n}_{n-2} = \boxed{n^{n-2}}.$$

Now, let us find the number of spanning trees for $K_{r,s}$. Let $K_{r,s}$ have an ordered vertex set $V(G) = \{u_1, \dots, u_r, w_1, \dots, w_s\}$. Each u_i has degree s and each w_j has degree r . The Laplacian matrix is:

$$L = \begin{pmatrix} sI_r & -\mathbf{1} \\ -\mathbf{1} & rI_s \end{pmatrix}$$

Delete the last row and column to obtain the $(r + s - 1) \times (r + s - 1)$ matrix:

$$L^* = \begin{pmatrix} sI_r & -\mathbf{1} \\ -\mathbf{1} & rI_{s-1} \end{pmatrix}$$

Now we perform the following row operations:

- For $i = 2, \dots, r$, perform $R_i \rightarrow R_i - R_1$. Rows $2, \dots, r$ now have $-s$ in column 1, s in column i , and zeros elsewhere (the right block cancels since the rows of $-\mathbf{1}$ are identical).
- For $j = r + 2, \dots, r + s - 1$, perform $R_j \rightarrow R_j - R_{r+1}$. Rows $r + 2, \dots, r + s - 1$ now have $-r$ in column $r + 1$, r in column j , and zeros elsewhere.
- Eliminate the -1 's in columns $2, \dots, r$ of row $r + 1$

$$R_{r+1} \rightarrow R_{r+1} + \frac{1}{s}(R_2 + \dots + R_r).$$

Row $r + 1$ becomes $(-r, 0, \dots, 0, r, 0, \dots, 0)$.

- Eliminate the -1 's in columns $r + 2, \dots, r + s - 1$ of row 1

$$R_1 \rightarrow R_1 + \frac{1}{r}(R_{r+2} + \dots + R_{r+s-1}).$$

Row 1 becomes $(s, 0, \dots, 0, -(s - 1), 0, \dots, 0)$.

After these operations, columns $2, \dots, r$ interact only with rows $2, \dots, r$, and columns $r + 2, \dots, r + s - 1$ interact only with rows $r + 2, \dots, r + s - 1$. Reordering so that indices $\{1, r + 1\}$ come first, the matrix takes block form

$$\begin{pmatrix} M & 0 \\ N & D \end{pmatrix},$$

where

$$M = \begin{pmatrix} s & -(s - 1) \\ -r & r \end{pmatrix}, \quad D = \begin{pmatrix} sI_{r-1} & 0 \\ 0 & rI_{s-2} \end{pmatrix}.$$

Therefore,

$$\begin{aligned} \tau(K_{r,s}) &= \det(L^*) = \det(M) \cdot \det(D) = (sr - r(s - 1)) \cdot s^{r-1} \cdot r^{s-2} = r \cdot s^{r-1} \cdot r^{s-2} \\ &= \boxed{r^{s-1} \cdot s^{r-1}}. \end{aligned}$$

□